

# Halcyon HA-10

Architecture and System Design Description

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Ten-seat hybrid-electric coastal amphibian. Fictional demonstration article — this aircraft is not any company's programme and no number in it is taken from, or intended to describe, any real aircraft.

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Roughly half of the network is water. The remainder is short paved strips of the kind found on small coastal fields. The aircraft is expected to alternate between the two within a single duty day, and the gear is selected for the surface on each approach rather than configured once for the day. That alternation is a normal operating condition, not an exception.

Ten occupants are carried, including two flight crew. The cabin is occupied throughout and is not a cargo hold; the occupant environment is therefore a function of the aircraft rather than a comfort feature of the interior.

Water operations are limited to a declared sea state. Outside that state the hull is outside its certified envelope, and the limitation is an operating limitation rather than a design margin.

Turnaround is measured in minutes at a water base with no line maintenance and no ground power. The aircraft is expected to be self-sufficient on the water: it starts, taxis, and departs on its own stored energy, and the operating concept assumes no external supply at the far end of a sector.

## 1.2 Configuration & Principal Characteristics

| Parameter                  | Value                                                                                       | Parameter                    | Value                                         |
|----------------------------|---------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------|
| Configuration              | High wing, twin wing-mounted electric propulsion units, T-tail, amphibious hull             | Maximum take-off weight      | 5,700 kg                                      |
| Occupants                  | 10 (2 flight crew + 8)                                                                      | Maximum landing weight       | 5,450 kg                                      |
| Wing span                  | 19.4 m                                                                                      | Operating empty weight       | approx. 3,650 kg                              |
| Overall length             | 14.2 m                                                                                      | Design payload               | 900 kg                                        |
| Hull                       | Single-step planing hull, two watertight compartments either side of a bulkhead at frame 14 | Energy store capacity        | 310 kWh usable (preliminary)                  |
| Powerplant                 | 2 × 340 kW electric propulsion units, wing-mounted, tractor                                 | Turbogenerator               | 1 × 400 kW class turbogenerator, aft fuselage |
| Propulsion architecture    | Hybrid-electric: turbogenerator and high-voltage store feed a common propulsion bus         | Fuel capacity                | 620 litres (two wing tanks)                   |
| Maximum operating altitude | FL120                                                                                       | Cruise altitude, typical     | 8,000 ft                                      |
| Design range               | 500 nm at maximum payload                                                                   | Landing distance requirement | 760 m paved; 900 m water run (preliminary)    |
| Landing gear               | Retractable tricycle, twin-wheel nose unit, single-wheel mains, water-capable bays          | Declared sea state           | Significant wave height 0.6 m (preliminary)   |

Figures given as preliminary are subject to change. The usable energy-store capacity, the water landing run and the declared sea state in particular are not frozen, and downstream work should cite them as preliminary rather than as fixed.

## 1.3 Certification & Safety Basis

The intended certification basis is 14 CFR Part 23, normal category, at Level 4 / high-speed as those levels are applied to a ten-occupant aeroplane, with the acceptable means of compliance for system safety taken from AC 23.1309-1E, Class III. Special conditions are anticipated in at least three areas: the high-voltage energy store and its installation, for which the existing Part 23 requirements were not written; the hull as a

primary structural and flotation element; and the amphibious gear as a configuration whose correct selection depends on the landing surface.

The severity scale used throughout the programme is the standard one — No Safety Effect, Minor, Major, Hazardous, Catastrophic — with the quantitative objectives of AC 23.1309-1E Class III applied to each. Development assurance is per ARP4754B and the safety assessment process per ARP4761A.

This document describes the design. It does not classify anything and it does not assess anything. The determination of what the aircraft is required to do, what happens when it fails to do it, how severe those outcomes are, and what must therefore be true of the architecture, is the work of the safety assessment and is not pre-empted here. That separation is deliberate and is discussed in Appendix A.

## 2 System Architecture Overview

### 2.1 System Inventory

Sixteen systems are defined at aircraft level. Five are treated as resources — they exist to supply other systems rather than to act on the aircraft themselves — and the distinction matters because a resource failure propagates to every consumer that stands on it. The thermal management loop is the clearest case on this aircraft: it is not a system anyone flies, and almost everything in the powertrain depends on it.

| Code | System                                 | Type     | Principal consumers or suppliers                               |
|------|----------------------------------------|----------|----------------------------------------------------------------|
| PRL  | Left Propulsion Unit                   | Function | Consumes the propulsion bus, TMS, FCA/FCB command              |
| PRR  | Right Propulsion Unit                  | Function | Consumes the propulsion bus, TMS, FCA/FCB command              |
| EST  | High-Voltage Energy Store              | Resource | Supplies the propulsion bus and both LV channels; consumes TMS |
| EGN  | Turbogenerator Set                     | Resource | Supplies the propulsion bus; consumes fuel, TMS                |
| TMS  | Thermal Management                     | Resource | Serves EST, EGN, PRL, PRR and the bus                          |
| FCA  | Flight Control Channel A               | Function | Consumes EL1; reads the primary display's sensors              |
| FCB  | Flight Control Channel B               | Function | Consumes EL2; reads the standby display's sensors              |
| EL1  | Electrical Distribution Channel 1      | Resource | Supplies FCA, DPP and the channel-1 load set                   |
| EL2  | Electrical Distribution Channel 2      | Resource | Supplies FCB, DPS and the channel-2 load set                   |
| DPP  | Primary Flight Display                 | Function | Consumes EL1; serves the crew and FCA                          |
| DPS  | Standby Display                        | Function | Consumes EL2 and a dedicated battery; serves the crew and FCB  |
| HUF  | Hull — Forward Compartment             | Function | Structural; interfaces LDG, ECS                                |
| HUA  | Hull — Aft Compartment & Water Systems | Function | Structural; hosts the water rudder                             |
| LDG  | Amphibious Landing Gear                | Function | Consumes EL1/EL2; interfaces HUF, HUA                          |
| FPR  | Fire & Thermal Protection              | Function | Consumes EL1/EL2; monitors EST, EGN, the cabin                 |
| ECS  | Cabin Environment                      | Function | Consumes EL1/EL2 and EGN bleed                                 |

### 2.2 System-to-System Interface Matrix

The matrix below records what each system needs from another in order to work. It is the starting point for understanding how a failure in one place is felt in another, and for testing whether two things claimed to be independent actually are.

| From           | To             | What crosses the interface                                                               |
|----------------|----------------|------------------------------------------------------------------------------------------|
| EST            | Propulsion bus | High-voltage DC through two string contactors; state of charge and thermal state as data |
| EGN            | Propulsion bus | High-voltage DC through the generator rectifier; available power as data                 |
| Propulsion bus | PRL / PRR      | High-voltage DC to each inverter — one electrical node, two consumers                    |
| EST            | EGN            | Starting current: the turbogenerator cannot start without the store                      |

| From      | To                    | What crosses the interface                                                                      |
|-----------|-----------------------|-------------------------------------------------------------------------------------------------|
| TMS       | EST / EGN / PRL / PRR | Coolant flow and return on a single loop with two pumps; coolant temperature as data            |
| EL1       | FCA / DPP             | 28 V DC to the channel-A computer, its actuator electronics, and the primary display            |
| EL2       | FCB / DPS             | 28 V DC to the channel-B computer, its actuator electronics, and the standby display            |
| EST       | EL1 / EL2             | Low-voltage supply through two independent DC/DC converters, one per channel                    |
| FCA / FCB | PRL / PRR             | Thrust command and thrust-lever position; each channel reaches both units                       |
| DPP       | FCA                   | Air data and inertial data from the primary sensor set                                          |
| DPS       | FCB                   | Air data and inertial data from the standby sensor set — a different source, deliberately       |
| LDG       | Multiple              | Weight-on-wheels and gear-position discretes to propulsion, displays and the environment system |
| HUA       | LDG                   | Water-rudder retraction interlock with the main gear selection                                  |
| FPR       | EST / EGN             | Thermal and smoke sensing in; extinguishant discharge and isolation command out                 |
| EGN       | ECS                   | Bleed air for cabin conditioning and for the windscreen anti-fog function                       |
| ECS       | HUF                   | Cabin pressure differential across the forward bulkhead as an interlock on the door             |

### 3 Zonal Breakdown

The aircraft is divided into ten zones for the purposes of installation and segregation. The hull compartments are the coarsest division and the most consequential: the watertight bulkhead at frame 14 is both a structural boundary and a zonal one, and nothing crosses it without being recorded in §4.

| Zone | Name                                 | Principal contents                                                            | Environment                                |
|------|--------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------|
| 110  | Nose and forward hull                | Nose gear bay, forward avionics rack, mooring gear                            | Unpressurised; wetted in water operations  |
| 120  | Flight deck                          | Displays, inceptors, standby display and its battery, thrust levers           | Pressurised, occupied                      |
| 200  | Forward cabin                        | Seats 1–4, forward ECS distribution, cabin fire detection                     | Pressurised, occupied                      |
| 300  | Aft cabin                            | Seats 5–8, aft ECS distribution, baggage                                      | Pressurised, occupied                      |
| 400  | Underfloor forward (fwd of frame 14) | Forward hull compartment, HV routing run A, bilge sensing                     | Unpressurised; watertight                  |
| 500  | Underfloor aft (aft of frame 14)     | Aft hull compartment, HV routing run B, water-rudder actuation                | Unpressurised; watertight                  |
| 600  | Energy store bay                     | High-voltage energy store, store contactors, TMS cold plate, containment vent | Unpressurised; fire zone; vented overboard |
| 700  | Left and right nacelles              | Propulsion units, inverters, TMS radiators, main gear bays                    | Unpressurised; fire zones                  |
| 800  | Aft fuselage                         | Turbogenerator, fuel gallery, rectifier, exhaust, TMS pumps                   | Unpressurised; fire zone                   |
| 900  | Wing and empennage                   | Wing tanks, control runs, water rudder, external lighting                     | Unpressurised                              |

### 4 Zone-Spanning Routings

Six routes cross a zone boundary and are recorded here because a hazard in one zone can reach another only along one of them.

R-1 High-voltage run A. Energy store bay (600) to the left nacelle (700L) through underfloor forward (400) and the left wing root. Carries propulsion-bus current at full power.

R-2 High-voltage run B. Energy store bay (600) to the right nacelle (700R) through underfloor aft (500) and the right wing root. Routed on the opposite side of the hull to R-1 and crossing the watertight bulkhead at a different frame.

R-3 Turbogenerator feed. Aft fuselage (800) to the energy store bay (600), carrying rectified high-voltage DC forward to the bus, and fuel aft from the wing gallery (900) to the burner.

R-4 Coolant ring. A single loop from the pumps in the aft fuselage (800) through the store bay (600), both nacelles (700L and 700R), and back. It passes through every zone that contains a heat source and is the only route that does so.

R-5 Channel 1 electrical run. Flight deck (120) to the left nacelle and the forward hull, on the left side of the aircraft throughout.

R-6 Channel 2 electrical run. Flight deck (120) to the right nacelle and the aft hull, on the right side of the aircraft throughout. R-5 and R-6 share no zone penetration except at the flight deck bulkhead, where they

pass through separate grommets 400 mm apart.

The energy store bay (600) is vented directly overboard and shares no ventilation path with the cabin. The bilge sensing in zones 400 and 500 is reported separately per compartment; there is no common bilge line between them, because a common line would defeat the bulkhead.

## 5 System Design Descriptions

The sixteen systems of §2.1 are described below in the same order and to the same six headings, so that any two can be compared without re-reading both in full.

### 5.1 Left Propulsion Unit (PRL)

#### 5.1.1 Purpose

The left propulsion unit converts electrical energy from the high-voltage propulsion bus into shaft power at the left propeller, and modulates that power in response to thrust commands from the flight control channels.

#### 5.1.2 Description & Architecture

The unit comprises a permanent-magnet motor, a three-phase inverter, a reduction gearbox and a constant-speed propeller. The inverter is the electrical boundary of the unit: upstream of it the aircraft is a high-voltage DC system, downstream of it a rotating machine.

The motor is wound as two independent three-phase sets on a shared rotor, each fed by its own half-bridge set within the inverter. Loss of one set leaves the unit able to produce reduced torque on the other; the two sets do not share gate drive, current sensing or control processing.

Propeller pitch is governed hydro-mechanically from an engine-driven pump within the gearbox, with electrical feather on command. Feathering does not depend on the propulsion bus.

#### 5.1.3 Redundancy & Reconfiguration

Two winding sets, two half-bridge sets, and an independent feather path. The unit is one of two on the aircraft and the sizing case for the pair is the continuation of a coastal sector after the loss of one; that case sets both the single-unit rating and the propeller pitch schedule, since a windmilling propeller on the failed side has to be feathered for the remaining unit's rating to be sufficient.

#### 5.1.4 Interfaces

In: high-voltage DC from the propulsion bus; coolant flow and return from the thermal management loop; thrust command and thrust-lever position from both flight control channels; low-voltage control power from channel 1.

Out: shaft power to the propeller; motor speed, torque, winding temperature and inverter state to the displays; a feather-complete discrete to the flight control channels.

#### 5.1.5 Installation & Segregation

The unit is installed in the left nacelle (zone 700L), which is a designated fire zone with its own detection loop and its own drainage. The inverter sits aft of the motor on the nacelle's rear bulkhead, cooled from the TMS radiator in the same nacelle. High-voltage run A enters the nacelle at the wing root and is the only high-voltage penetration on this side.

#### 5.1.6 Operation

The crew commands thrust with a single lever per side. Above a defined coolant temperature the unit derates automatically and the derate is annunciated. Feather is available as a guarded switch on the overhead panel and is also commanded automatically when the flight control channels detect a locked rotor.

### 5.2 Right Propulsion Unit (PRR)

#### 5.2.1 Purpose

The right propulsion unit converts electrical energy from the high-voltage propulsion bus into shaft power at the right propeller, and modulates that power in response to thrust commands from the flight control

channels.

### 5.2.2 Description & Architecture

The right unit is the mirror of the left and shares its part numbers: the same motor with two independent winding sets, the same two-set inverter, the same gearbox and the same constant-speed propeller.

That commonality is deliberate — it is a maintenance and spares decision taken at programme level — and it is recorded here rather than buried, because two units of identical design are not two independent units in every respect. What differs between them is installation: the side of the aircraft, the high-voltage run that feeds them, and the low-voltage channel that controls them.

### 5.2.3 Redundancy & Reconfiguration

As the left unit: two winding sets, two half-bridge sets, an independent feather path, and a pair-level case in which one unit continues the sector alone. Nothing in the right unit reconfigures to cover a left-unit failure other than by carrying more of the load within its own rating.

### 5.2.4 Interfaces

In: high-voltage DC from the propulsion bus; coolant flow and return from the thermal management loop; thrust command and thrust-lever position from both flight control channels; low-voltage control power from channel 2.

Out: shaft power to the propeller; motor speed, torque, winding temperature and inverter state to the displays; a feather-complete discrete to the flight control channels.

### 5.2.5 Installation & Segregation

The unit is installed in the right nacelle (zone 700R), a designated fire zone with its own detection loop and drainage. It is fed by high-voltage run B, which crosses the watertight bulkhead at a different frame from run A and travels on the opposite side of the hull. Control power comes from channel 2, so the two units do not share a low-voltage source.

### 5.2.6 Operation

As the left unit. The two thrust levers are mechanically independent and are not ganged; there is no single lever that commands both.

## 5.3 High-Voltage Energy Store (EST)

### 5.3.1 Purpose

The energy store holds the propulsion energy for the sector and delivers it to the high-voltage propulsion bus, within the limits of its declared thermal and state-of-charge envelope.

### 5.3.2 Description & Architecture

The store is built from modules arranged in two parallel strings, each with its own contactor pair, its own current sensing and its own module-level monitoring. A battery management system supervises cell voltage, module temperature and string current, and is the authority that opens a contactor.

Each string can be isolated at its contactors without disturbing the other. The strings share the enclosure, the cold plate and the vent path; they do not share monitoring hardware, contactors or wiring above the module level.

The enclosure is a sealed steel case with a burst membrane venting directly overboard through the underside of the store bay. It carries no ventilation path to the cabin.

The propulsion bus itself is part of this installation: a potted busbar assembly on the forward wall of the same bay, carrying four contactors — one per energy source and one per propulsion unit. It is a bus and not a ring,

and that is the architectural fact from which most of its behaviour follows: every source and every consumer meets at one electrical node. Each contactor has its own controller and reports its own state; no two share a controller. An insulation-monitoring device measures leakage to airframe continuously and reports it as data.

### 5.3.3 Redundancy & Reconfiguration

Two strings, independently isolable, either of which can feed the bus at reduced power. The store is one of two energy sources on the aircraft; the turbogenerator is the other, and the two are dissimilar by design rather than by installation. The reserve case that sizes the store is a single go-around at the destination followed by a diversion to an alternate; that case is preliminary and is expected to move.

### 5.3.4 Interfaces

In: coolant flow and return from the thermal management loop; low-voltage control power, taken from both channels through a diode-OR so that the management system does not depend on either alone; charging current on the ground through a dedicated inlet.

Out: high-voltage DC to the propulsion bus through the string contactors; low-voltage supply to both electrical channels through two independent DC/DC converters; state of charge, module temperatures, string currents and contactor state to the displays; a thermal-event discrete to the fire and thermal protection system.

### 5.3.5 Installation & Segregation

The store occupies its own bay (zone 600) below the cabin floor and aft of the forward watertight bulkhead. The bay is a designated fire zone, is vented overboard, and shares no wall with an occupied compartment other than the cabin floor, which is an insulated double skin. The two strings sit either side of a central web within the enclosure.

### 5.3.6 Operation

The store is charged on the ground through the dedicated inlet, and is available from battery-master selection. In flight the crew sees state of charge, available power and the warmest module. There is no crew action that opens a contactor directly; isolation is a management-system function, and the crew's control is a master switch that requests it.

## 5.4 Turbogenerator Set (EGN)

### 5.4.1 Purpose

The turbogenerator converts fuel into electrical energy in flight and delivers it to the high-voltage propulsion bus, supplementing or replacing the energy store as the sector requires.

### 5.4.2 Description & Architecture

The set is a single-shaft gas turbine driving a generator through a reduction gearbox, with a rectifier converting its output to high-voltage DC at the bus. It is a single unit; there is no second turbogenerator.

Fuel is held in two integral wing tanks, one per side, feeding a single gallery aft to the burner. Each tank has its own boost pump — left on channel 1, right on channel 2 — and its own shutoff valve, with a crossfeed cock so either tank can supply the set. Contents are measured by capacitance probes with a separate low-level float switch per tank, two dissimilar means. There is no fuel in the hull and none is jettisonable.

Starting is from the energy store, which means the store is a prerequisite for the generator and not merely an alternative to it — a dependency recorded here because it is the kind of thing an architecture diagram hides.

The rectifier is a passive unit with no control processing; the generator's output is regulated at the machine.

### 5.4.3 Redundancy & Reconfiguration

None within the set: it is a simplex system. Its redundancy partner is the energy store, and the pair is dissimilar — one stores, one generates, and they fail for unrelated reasons. The set can be shut down and its bus contactor opened without affecting the store's path to the bus.

#### 5.4.4 Interfaces

In: coolant flow and return from the thermal management loop; low-voltage control power from channel 2; power to the boost pumps, split between channels; a start command and starting current from the energy store; bleed offtake demand from the cabin environment system.

Out: high-voltage DC to the propulsion bus through the rectifier and its contactor; bleed air to the cabin environment system; shaft speed, exhaust gas temperature, fuel flow and available power to the displays; a thermal and overheat discrete to the fire and thermal protection system.

#### 5.4.5 Installation & Segregation

The set is installed in the aft fuselage (zone 800), a designated fire zone with its own detection loop, its own extinguishant bottle and a drained, ventilated compartment. The exhaust exits above the waterline on the port side. Tanks are within the wing box (zone 900), outboard of the hull and above the waterline in all normal attitudes; the gallery enters the fire zone on route R-3 through a firewall fitting with a shutoff valve immediately outside it. No fuel line runs through the cabin, the store bay or either nacelle. The rectifier sits forward of the firewall.

#### 5.4.6 Operation

The set is started after the store is live and is normally run for the whole sector. The crew selects it on or off; there is no partial mode. Shutdown in flight leaves the aircraft on stored energy, which is a normal reversion and is annunciated as such rather than as a failure.

### 5.5 Thermal Management (TMS)

#### 5.5.1 Purpose

The thermal management system removes heat from the energy store, the turbogenerator, both propulsion units and the propulsion bus, and holds each within its declared operating temperature range.

#### 5.5.2 Description & Architecture

A single liquid loop serves every heat source on the aircraft. Two pumps in the aft fuselage circulate coolant through a ring that reaches the store bay, both nacelles and back; each nacelle carries a radiator with its own controllable flap.

The loop is one loop. It has two pumps, two radiators and two flap controllers, but a single body of coolant and a single set of pipework, and a loss of containment anywhere in it is a loss of the whole loop. This is the aircraft's most widely shared resource and the section is written to say so rather than to imply otherwise.

Cold plates at the store and the busbar are integral to those assemblies. The propulsion inverters and the generator rectifier are plate-cooled from the same loop.

#### 5.5.3 Redundancy & Reconfiguration

Two pumps on separate electrical channels, either sufficient at reduced flow; two radiators, either sufficient at reduced ambient. No redundancy in the coolant path itself. Consumers derate on rising coolant temperature — the store first, then the propulsion units — and the derate order is a design choice recorded here because it determines what the aircraft still does as the loop degrades.

#### 5.5.4 Interfaces

In: low-voltage power to pump 1 from channel 1 and pump 2 from channel 2; flap commands from the same channels; coolant temperature and pressure sensing at four points in the ring.

Out: coolant flow and return to the energy store, the propulsion bus, both propulsion units and the turbogenerator; loop temperature, pressure and pump state to the displays; a derate request to each consumer.

### 5.5.5 Installation & Segregation

Pumps and the expansion tank are in the aft fuselage (zone 800). The ring follows route R-4 and is the only route that enters every zone containing a heat source. It crosses the watertight bulkhead once, at frame 14, through a sealed penetration shared with no other service.

### 5.5.6 Operation

The system runs automatically whenever the aircraft is live. The crew sees loop temperature and pump state, and has a pump-select switch. There is no crew action that reconfigures the coolant path, because there is no alternative path to select.

## 5.6 Flight Control Channel A (FCA)

### 5.6.1 Purpose

Channel A commands the positions of the primary control surfaces in response to pilot and autopilot inputs, and monitors the response of those surfaces against the command it issued.

### 5.6.2 Description & Architecture

Channel A is a flight control computer, its actuator control electronics, and its own sensor set. Primary control is fly-by-wire to electro-hydrostatic actuators, one per surface half: left and right elevator, left and right aileron, and the upper rudder segment.

The channel hosts software developed by one of two teams working to a dissimilarity argument; channel B hosts the other. The two channels use different processors and different air-data sources, and they do not exchange state except through a monitored cross-compare of surface position.

Secondary control — flap drive and pitch trim — is commanded by whichever channel is in control, through a single power drive unit per function with asymmetry braking at each flap section.

### 5.6.3 Redundancy & Reconfiguration

Channel A is one of two. Each channel alone can fly the aircraft through its own half of each surface pair; the surfaces are sized so that one half of each pair is sufficient. Transfer between channels is automatic on a monitored disagreement and is annunciated. A mechanical reversion path exists to pitch trim and to the lower rudder segment only, and it is not a selectable mode — it is what remains when both channels are gone.

### 5.6.4 Interfaces

In: 28 V DC from electrical channel 1; air data and inertial data from the primary display's sensor set; pilot inceptor position; autopilot commands; surface position feedback from its own actuators; thrust-lever position.

Out: actuator commands to its half of each surface pair; thrust commands to both propulsion units; control law state, surface positions and channel health to the displays; a cross-compare word to channel B.

### 5.6.5 Installation & Segregation

The channel A computer is installed in the forward avionics rack on the left side of the nose (zone 110). Its wiring follows route R-5 on the left of the aircraft throughout. It is powered from electrical channel 1 and shares no power, sensing or processing element with channel B; the two share only the airframe and the surfaces they drive.

### 5.6.6 Operation

The crew flies through conventional inceptors with envelope protection active in the normal control law. Degraded laws are annunciated. Channel selection is automatic; the crew has a channel-select switch for maintenance and for a deliberate transfer.

## 5.7 Flight Control Channel B (FCB)

### 5.7.1 Purpose

Channel B commands the positions of the primary control surfaces in response to pilot and autopilot inputs, and monitors the response of those surfaces against the command it issued.

### 5.7.2 Description & Architecture

Channel B is the dissimilar counterpart of channel A: a different processor family, software from the second development team, and its own sensor set taken from the standby display's sources rather than the primary's.

It drives the other half of each surface pair — right and left elevator inboard sections, the opposite aileron actuator, and the lower rudder segment. The lower rudder segment is the one that retains a mechanical path, so channel B's surface set and the mechanical reversion set overlap by design.

### 5.7.3 Redundancy & Reconfiguration

As channel A, and symmetrically: either channel alone is sufficient, transfer is automatic on a monitored disagreement, and the mechanical path is what remains below both. The asymmetry worth recording is the sensor source — channel B reads the standby display's air data, which is itself on a separate supply — so the two channels do not fail together on a common air-data source.

### 5.7.4 Interfaces

In: 28 V DC from electrical channel 2; air data and inertial data from the standby display's sensor set; pilot inceptor position; autopilot commands; surface position feedback from its own actuators; thrust-lever position.

Out: actuator commands to its half of each surface pair; thrust commands to both propulsion units; control law state, surface positions and channel health to the displays; a cross-compare word to channel A.

### 5.7.5 Installation & Segregation

The channel B computer is installed in the aft cabin equipment bay on the right side (zone 300), not in the forward rack. Its wiring follows route R-6 on the right of the aircraft throughout. The physical separation between the two computers is the length of the cabin, and it is the basis on which the two are installed rather than co-located.

### 5.7.6 Operation

As channel A. The channel-select switch selects either channel explicitly; there is no position in which both are commanded to drive the same surface half.

## 5.8 Electrical Distribution Channel 1 (EL1)

### 5.8.1 Purpose

Channel 1 converts high-voltage energy to low voltage and distributes it to the channel-1 loads, including flight control channel A and the primary display.

### 5.8.2 Description & Architecture

A DC/DC converter at the energy store steps high voltage down to 28 V and feeds a channel-1 busbar in the forward avionics rack. Distribution is through solid-state power controllers, one per load, each individually resettable and each reporting its own state.

A channel-1 battery sits across the bus and carries the channel through a converter failure or a high-voltage isolation, for a duration that the assessment will size rather than this document.

There is no bus tie between channel 1 and channel 2. The absence is deliberate: a tie would make a fault on one bus reachable from the other, and the two channels exist to not do that.

### 5.8.3 Redundancy & Reconfiguration

Channel 1 is one of two. Every flight-critical load set is carried by one channel or the other, and the split is by function rather than by convenience — channel A of the flight controls, the primary display, the left boost pump, coolant pump 1. There is no reconfiguration between channels; the redundancy is at the load, not at the bus.

### 5.8.4 Interfaces

In: high-voltage DC from the energy store through the channel-1 DC/DC converter; charging current to the channel-1 battery.

Out: 28 V DC to flight control channel A, the primary display, the left fuel boost pump, coolant pump 1, the channel-1 half of the gear control, and the channel-1 fire detection loop; bus voltage, load states and battery state to the displays.

### 5.8.5 Installation & Segregation

The channel-1 converter is at the energy store bay (zone 600); the channel-1 busbar and battery are in the forward avionics rack (zone 110). Wiring follows route R-5 on the left of the aircraft. Channel 1 and channel 2 share no generation, conversion or busbar element, and cross only at the flight deck bulkhead, through separate grommets 400 mm apart.

### 5.8.6 Operation

The channel is energised by the battery master. Individual loads can be reset from the overhead panel. The crew has no control that connects channel 1 to channel 2.

## 5.9 Electrical Distribution Channel 2 (EL2)

### 5.9.1 Purpose

Channel 2 converts high-voltage energy to low voltage and distributes it to the channel-2 loads, including flight control channel B and the standby display.

### 5.9.2 Description & Architecture

Channel 2 mirrors channel 1: its own DC/DC converter at the energy store, its own busbar, its own solid-state power controllers, and its own battery.

Its busbar is in the aft cabin equipment bay rather than the forward rack, so the two channels are separated along the length of the aircraft as well as across it. The standby display carries a further dedicated battery of its own, independent of the channel-2 battery — that display is the only load on the aircraft with two levels of electrical backing.

### 5.9.3 Redundancy & Reconfiguration

As channel 1, symmetrically, and with the same deliberate absence of a bus tie. The channel-2 load set is flight control channel B, the standby display, the right boost pump, coolant pump 2, the channel-2 half of the gear control, and the channel-2 fire detection loop.

### 5.9.4 Interfaces

In: high-voltage DC from the energy store through the channel-2 DC/DC converter; charging current to the channel-2 battery and to the standby display's dedicated battery.

Out: 28 V DC to the loads listed above; bus voltage, load states and battery states to the displays.

#### 5.9.5 Installation & Segregation

The channel-2 converter is at the energy store bay (zone 600) on the opposite face of the enclosure to channel 1's. The channel-2 busbar and battery are in the aft cabin equipment bay (zone 300). Wiring follows route R-6 on the right of the aircraft.

#### 5.9.6 Operation

As channel 1. Both channels come up on the battery master and neither is selectable independently in normal operation.

### 5.10 Primary Flight Display (DPP)

#### 5.10.1 Purpose

The primary flight display presents attitude, air data, powertrain state and warnings to the flight crew, and supplies the sensed data that flight control channel A uses.

#### 5.10.2 Description & Architecture

Two identical display units, one per pilot, driven from a common primary sensor set: an air data and inertial reference unit, a magnetometer and a pitot-static system with two heated probes.

The powertrain page is particular to this aircraft. It shows state of charge, available power from each source, coolant loop temperature and the warmest energy-store module, because on a hybrid aircraft the question "how much flying is left in it" is not answered by a fuel gauge.

Warnings are generated by a crew alerting function hosted in the display units and are presented in three levels with distinct aural signatures. The alerting function reads discretes and data from every system in §2.2 and is the single place where they are prioritised.

#### 5.10.3 Redundancy & Reconfiguration

Two display units on one sensor set, and a standby display on an entirely separate one. Loss of one display unit leaves the other presenting the full picture; loss of the primary sensor set leaves the standby display and channel B, which reads it. Cross-comparison between the primary and standby sources is continuous and a disagreement is annunciated rather than resolved automatically — the design does not vote two sources.

#### 5.10.4 Interfaces

In: 28 V DC from electrical channel 1; pitot, static and total air temperature; inertial and magnetic heading; data and discretes from every system listed in §2.2.

Out: air data and inertial data to flight control channel A; the displayed picture and the aural alerting to the crew; a cross-compare word to the standby display.

#### 5.10.5 Installation & Segregation

Both units are in the flight deck panel (zone 120). The primary sensor set is in the forward avionics rack (zone 110) with the channel A computer, and the two share route R-5 and electrical channel 1 — a commonality recorded here as an installation item to be examined, since the primary display and channel A are the two things that depend on it.

#### 5.10.6 Operation

The displays come up with the battery master. Either pilot's unit can be selected to show either the flight page or the powertrain page. Alerting cannot be inhibited in flight other than by the standard master-caution acknowledgement.

## 5.11 Standby Display (DPS)

### 5.11.1 Purpose

The standby display presents attitude and air data on a source and a power supply independent of the primary display, and supplies the sensed data that flight control channel B uses.

### 5.11.2 Description & Architecture

A single integrated standby unit with its own solid-state attitude sensor, its own static and pitot connection to the second probe, and its own battery.

It is the only load on the aircraft with two levels of electrical backing: electrical channel 2, and behind that a dedicated battery in the flight deck. That arrangement exists because the standby display is also a sensor — flight control channel B reads it — so its loss is not only a loss of information to the crew.

### 5.11.3 Redundancy & Reconfiguration

Not redundant in itself; it is the redundancy. Its independence from the primary display is across four things at once: sensor, probe, electrical channel and battery. The design deliberately does not tie it to the primary picture in any way that would let one correct the other.

### 5.11.4 Interfaces

In: 28 V DC from electrical channel 2, with automatic transfer to its dedicated battery; pitot and static from the second probe.

Out: attitude and air data to flight control channel B; the standby picture to the crew; a cross-compare word to the primary display.

### 5.11.5 Installation & Segregation

The unit and its battery are in the flight deck (zone 120) on the right of the panel. Its wiring follows route R-6. It shares no probe, no sensor, no converter and no busbar with the primary display.

### 5.11.6 Operation

The display is always on when the aircraft is live and has no crew controls beyond brightness and a barometric setting. Transfer to its own battery is automatic and annunciated.

## 5.12 Forward Hull Compartment (HUF)

### 5.12.1 Purpose

The forward hull compartment carries the water and landing loads forward of the bulkhead at frame 14 and retains watertight integrity in that region.

### 5.12.2 Description & Architecture

A single-step planing hull of aluminium construction, framed at 500 mm and divided from the aft compartment by a watertight bulkhead at frame 14. The forward compartment contains the nose gear bay, the forward avionics rack above the waterline, and the mooring gear.

The nose gear bay is the only opening in the compartment below the waterline; it is sealed by the gear doors and drained through a one-way valve above the static waterline. Bilge level is sensed in this compartment separately from the aft one, and there is no bilge line between the two — a common line would defeat the bulkhead, which is the point of having it.

The step is at frame 11, forward of the bulkhead.

### 5.12.3 Redundancy & Reconfiguration

The bulkhead is the redundancy: the hull is designed so that flooding of either compartment alone leaves the other providing buoyancy. There is nothing to reconfigure — this is structure, and its behaviour is set at build.

#### 5.12.4 Interfaces

In: gear door position and nose gear position from the landing gear system; cabin pressure differential from the environment system as an interlock on the forward door.

Out: bilge level in this compartment to the displays; structural load path to the nose gear and the mooring points.

#### 5.12.5 Installation & Segregation

Zones 110 and 400. High-voltage run A crosses the bulkhead at frame 14 through a sealed penetration; the coolant ring crosses at the same frame through a separate one. Those two, and the channel-1 electrical run, are the only services that cross, and each is individually sealed.

#### 5.12.6 Operation

No crew operation. Bilge level is presented on the ground and in flight; the crew's action on a bilge indication is a procedure, not a system function.

### 5.13 Aft Hull Compartment & Water Systems (HUA)

#### 5.13.1 Purpose

The aft hull compartment carries the water and landing loads aft of the bulkhead at frame 14, retains watertight integrity in that region, and provides directional control on the water.

#### 5.13.2 Description & Architecture

The aft compartment mirrors the forward one in construction and contains the main gear bays, the aft cabin equipment bay above the waterline, and the water rudder.

The water rudder is a retractable blade at the sternpost, deployed for water taxi and retracted for take-off, landing and all land operations. It is driven by its own electro-mechanical actuator and steered from the rudder pedals through a mechanical path that is independent of the air rudder control run — the two share the pedals and nothing else.

Retraction is interlocked with main gear selection: selecting gear down for a land landing retracts the water rudder, and the interlock is a hardwired discrete rather than a software function.

#### 5.13.3 Redundancy & Reconfiguration

The bulkhead again, as for the forward compartment. The water rudder is simplex; with its actuator failed it can be retracted by a manual release in the aft cabin, and directional control on the water reverts to differential thrust.

#### 5.13.4 Interfaces

In: 28 V DC from both channels to the water-rudder actuator through a diode-OR; rudder pedal position mechanically; main gear selection as a hardwired retraction interlock.

Out: water-rudder position and bilge level in this compartment to the displays; structural load path to the main gear and the empennage.

#### 5.13.5 Installation & Segregation

Zones 500 and 900. High-voltage run B crosses the bulkhead at a different frame from run A. The water-rudder actuator sits above the static waterline in the sternpost; its mechanical run to the pedals is on the right of the hull, opposite the air rudder run.

### 5.13.6 Operation

The crew deploys and retracts the water rudder with a guarded switch, and the retraction interlock overrides the switch when the main gear is selected down. Position is annunciated.

## 5.14 Amphibious Landing Gear (LDG)

### 5.14.1 Purpose

The landing gear extends, retracts and indicates a retractable tricycle gear suitable for paved surfaces, and keeps it stowed and sealed for water operations.

### 5.14.2 Description & Architecture

A twin-wheel nose unit and single-wheel main units, all electro-mechanically actuated with no hydraulic system on the aircraft. Each unit has its own actuator and its own uplock and downlock, and each lock has two independent position sensors.

Gear position is therefore sensed by two dissimilar means per unit, and the indication the crew sees is a comparison of the two rather than a single sensor's word. On an amphibian, gear position is not a routine indication: the same configuration that is correct for a runway is wrong for water, and the design treats the indication accordingly.

An emergency extension releases the uplocks mechanically and lets the units free-fall to a locked position.

### 5.14.3 Redundancy & Reconfiguration

Two independent sensors per lock; power to the actuators split between the two electrical channels so that no single channel loss stops the gear; a mechanical free-fall extension below all of that. There is no redundancy in the actuator per unit.

### 5.14.4 Interfaces

In: 28 V DC from both channels, split per unit; gear selection from the flight deck lever; the selected landing surface from the crew's approach configuration.

Out: gear and lock position from both sensor sets to the displays and the crew alerting function; weight-on-wheels discretes to propulsion, the environment system and the displays; a retraction interlock to the water rudder.

### 5.14.5 Installation & Segregation

The nose unit is in zone 110, the main units in the nacelle bays (zone 700). All three bays are sealed against water ingress by their doors and drained above the static waterline. Channel-1 power reaches the nose and left main; channel-2 power the right main.

### 5.14.6 Operation

The crew selects gear up or down with a conventional lever. The alerting function compares gear position against the selected landing surface and annunciates a mismatch; that comparison is the reason the surface selection is an explicit crew input rather than inferred.

## 5.15 Fire & Thermal Protection (FPR)

### 5.15.1 Purpose

The fire and thermal protection system detects fire, overheat and thermal-runaway conditions in the designated fire zones and in the cabin, contains them where containment is designed in, and gives the crew the means to isolate and to discharge.

### 5.15.2 Description & Architecture

Four detection zones: the energy store bay, the turbogenerator compartment, and each of the two nacelles. Each has a dual-loop detector — two loops on separate electrical channels, in an AND arrangement for warning and an OR arrangement for a loop fault, so a single loop failure produces a fault indication and not a fire warning.

The energy store bay is different in kind from the other three. It carries module-level thermal sensing reported by the battery management system in addition to the loop detectors, and its response is containment rather than extinguishing: the enclosure and its overboard vent are designed to hold a single-module event inside the case and route its products outside the aircraft. There is no extinguishant in the store bay, because there is no extinguishant that usefully addresses a cell in thermal runaway.

The turbogenerator compartment carries a single extinguishant bottle. The nacelles carry none; their response is isolation — opening the propulsion-unit contactor and feathering.

Cabin and baggage smoke detection is separate, photoelectric, and reports to the crew alerting function.

### 5.15.3 Redundancy & Reconfiguration

Dual loops per zone on separate channels; two independent sensing principles in the store bay (loop detection and module thermal sensing); a single bottle in the generator compartment with no reserve. Detection is redundant; suppression is not, and the design answers that by making containment and isolation the primary responses and extinguishant the exception.

### 5.15.4 Interfaces

In: 28 V DC from both channels, one per loop; module temperatures and thermal-event discretes from the energy store; overheat discretes from the turbogenerator; smoke detection from the cabin and baggage.

Out: fire and overheat warnings to the crew alerting function; a fuel shutoff command to the turbogenerator feed; a contactor-open command to the propulsion bus; an extinguishant discharge command to the generator bottle; the store containment vent state to the displays.

### 5.15.5 Installation & Segregation

Loops are routed within their own zones and leave them on the electrical run of their own channel — loop 1 on route R-5, loop 2 on route R-6 — so that a single route does not carry both loops of any zone. The store bay vent discharges through the underside of the hull, outboard of the bilge and clear of the cabin air inlet.

### 5.15.6 Operation

Detection is continuous and needs no crew selection. Discharge in the generator compartment is a guarded two-action control. In the store bay there is no discharge control; the crew actions on a thermal event are isolation and a procedure, and the system's role is to tell them and to contain what it can.

## 5.16 Cabin Environment (ECS)

### 5.16.1 Purpose

The cabin environment system maintains a habitable cabin — pressure, temperature and ventilation — throughout the mission, and clears the windscreen.

### 5.16.2 Description & Architecture

Conditioning air is bled from the turbogenerator, cooled through an air-cycle pack in the aft fuselage, and distributed forward through overhead ducting to the flight deck and both cabin zones. Cabin pressure is controlled by a single outflow valve at the aft pressure bulkhead with an independent positive-pressure relief valve beside it.

Because the turbogenerator is the only bleed source, the pack has an electrical alternative: a recirculation fan and an electric heater on channel 2 provide ventilation and heating with the generator shut down, without

pressurisation. Flying on stored energy alone is therefore an unpressurised condition by design, which is consistent with a maximum operating altitude of FL120.

Windscreen anti-fog is bleed-fed with an electric backup element.

### **5.16.3 Redundancy & Reconfiguration**

One pack, one outflow valve, one relief valve; an electrical ventilation and heating path behind the pack; and an altitude limit that makes the unpressurised case an operating condition rather than an emergency. There is no second pack.

### **5.16.4 Interfaces**

In: bleed air from the turbogenerator; 28 V DC from both channels to the pack controller, the recirculation fan and the heater; weight-on-wheels from the landing gear; cabin altitude and differential sensing.

Out: conditioned air to the flight deck and cabin; cabin differential pressure to the forward hull door interlock and to the displays; pack and valve state to the displays; the anti-fog supply to the windscreen.

### **5.16.5 Installation & Segregation**

The pack is in the aft fuselage (zone 800) forward of the turbogenerator firewall. The outflow and relief valves are at the aft pressure bulkhead. The cabin air inlet is on the upper fuselage, well clear of the energy store's overboard vent on the underside — a separation recorded here because the two would otherwise be the obvious pairing to examine.

### **5.16.6 Operation**

The system runs automatically. The crew selects cabin temperature and has a manual outflow-valve override. On generator shutdown the system reverts to recirculation and the reversion is annunciated as a change of state rather than as a failure.

## 6 Zonal Model & System Schematics

The zonal model is the ten-zone division of §3 applied to the physical aircraft: each zone is a volume with a boundary, an environment and a list of what is installed in it. The schematics below are per-system views drawn on that model, so that a system can be read either as a functional chain or as a set of things in places.

Each figure carries the same three layers: the system's own components, the zones they sit in, and the routes of §4 that connect them. Where a system crosses the watertight bulkhead at frame 14, the crossing is marked, because that boundary is the one the hull design depends on.

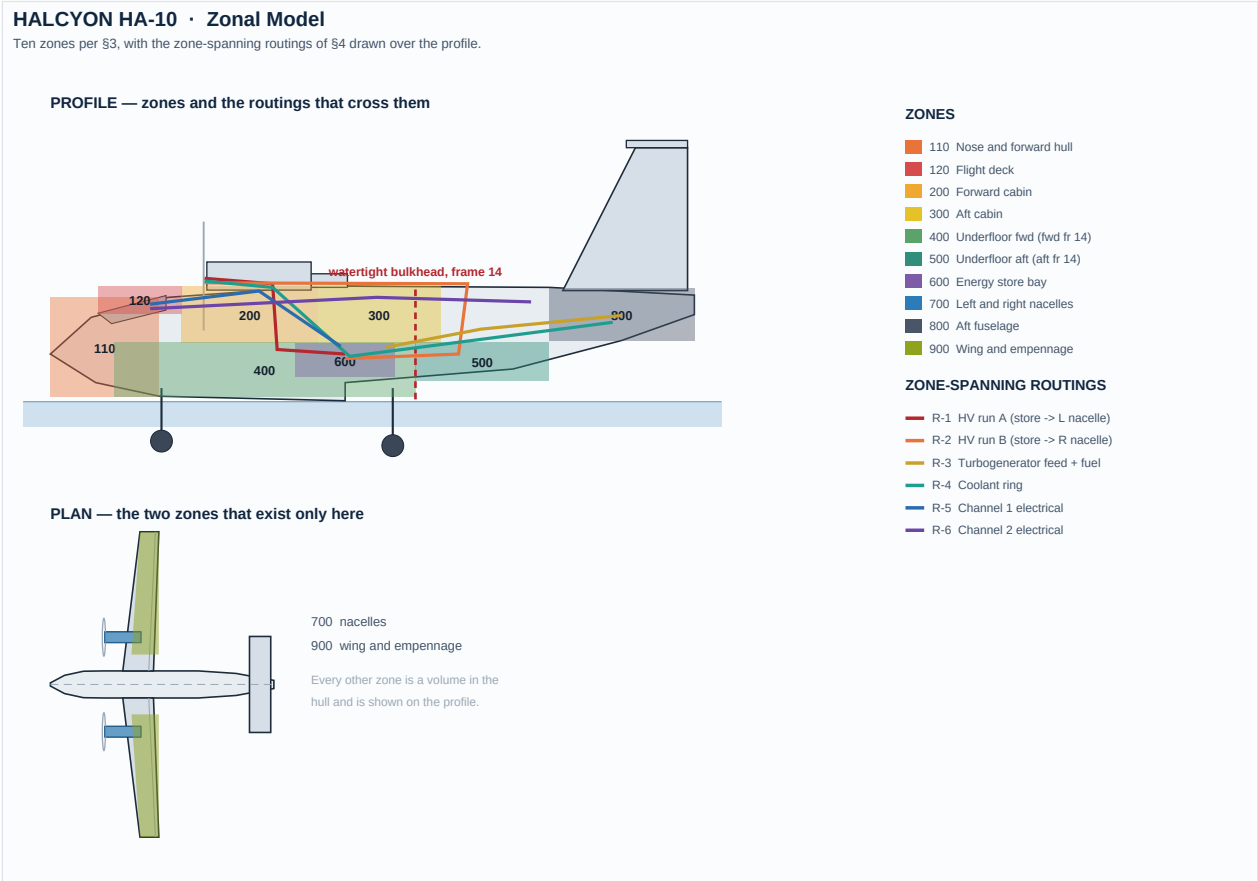


Figure 6-1 Zonal model. Ten zones per §3, with the zone-spanning routings of §4 drawn over the profile and the watertight bulkhead at frame 14 marked.

### 6.1 Left Propulsion Unit (PRL)

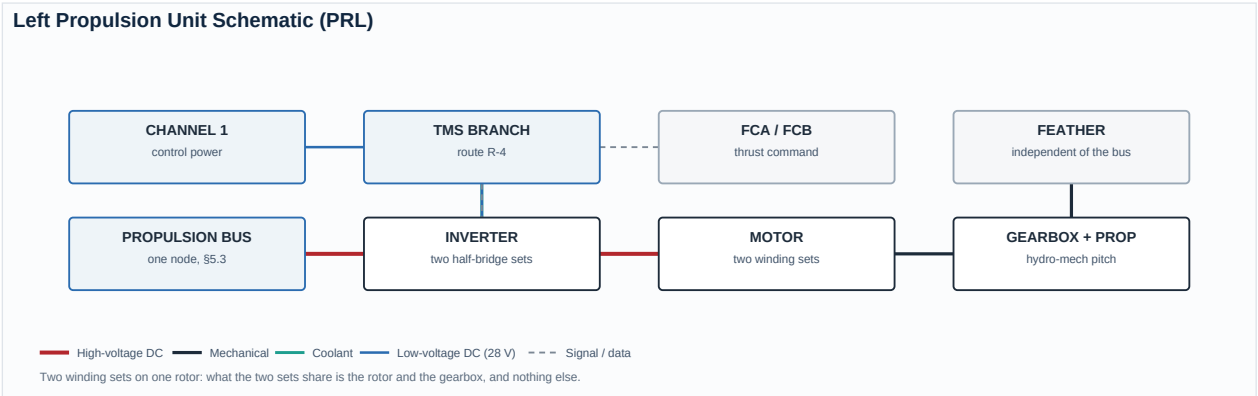


Figure 6-2 Left Propulsion Unit (PRL). Single-line diagram from the propulsion bus through the run A penetration to the inverter, motor and gearbox, with the two winding sets shown as separate paths from the two half-bridge sets. The coolant

branch from route R-4 and the channel-1 control feed are shown entering the nacelle separately. Read off it: what is common between the two winding sets, which is the rotor and the gearbox and nothing else.

6.2 Right Propulsion Unit (PRR)

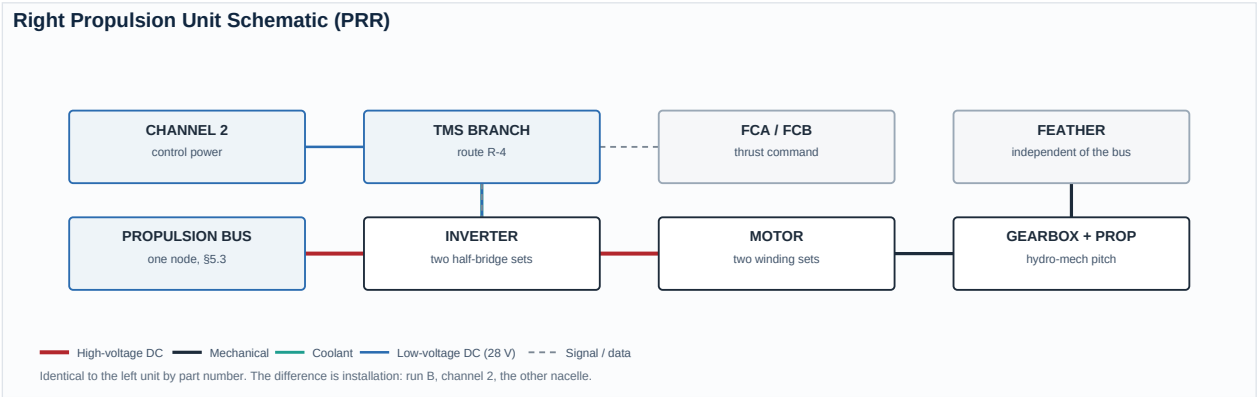


Figure 6-3 Right Propulsion Unit (PRR). The mirror of Figure 6-1, on run B and channel 2, drawn to the same key so the two can be laid side by side. Read off it: the two units are identical in design and differ only in installation, which is the distinction the independence argument turns on.

6.3 High-Voltage Energy Store (EST)

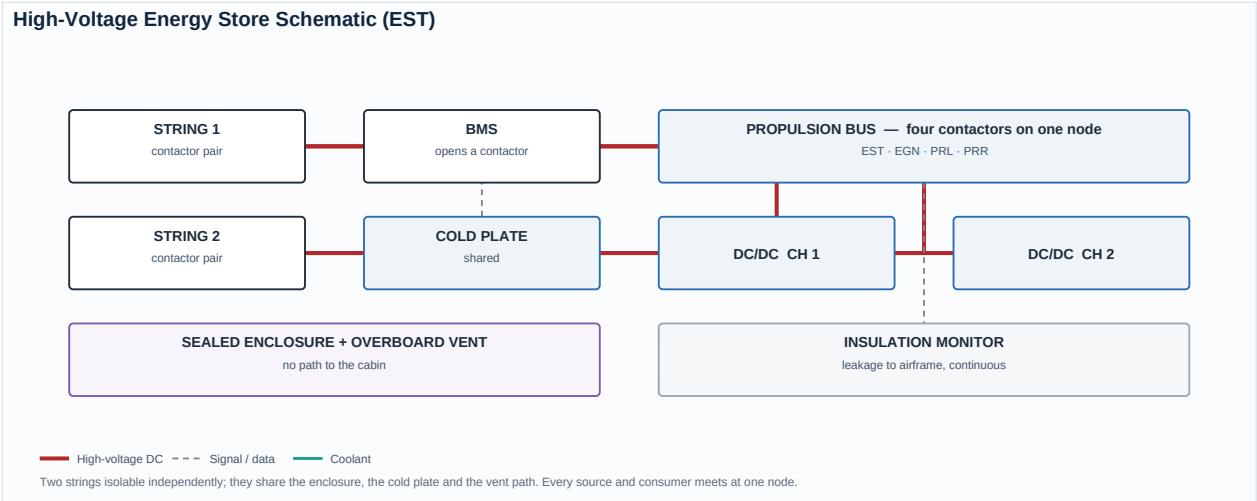


Figure 6-4 High-Voltage Energy Store (EST). The store enclosure with two strings, their contactor pairs and the battery management system, and the busbar assembly on the forward wall of the same bay with its four contactors. The overboard vent path and the cold-plate connection to route R-4 are shown leaving the bay. Read off it: every source and every consumer of high-voltage power meets at one node, and that node is in the same bay as the cells.

6.4 Turbogenerator Set (EGN)

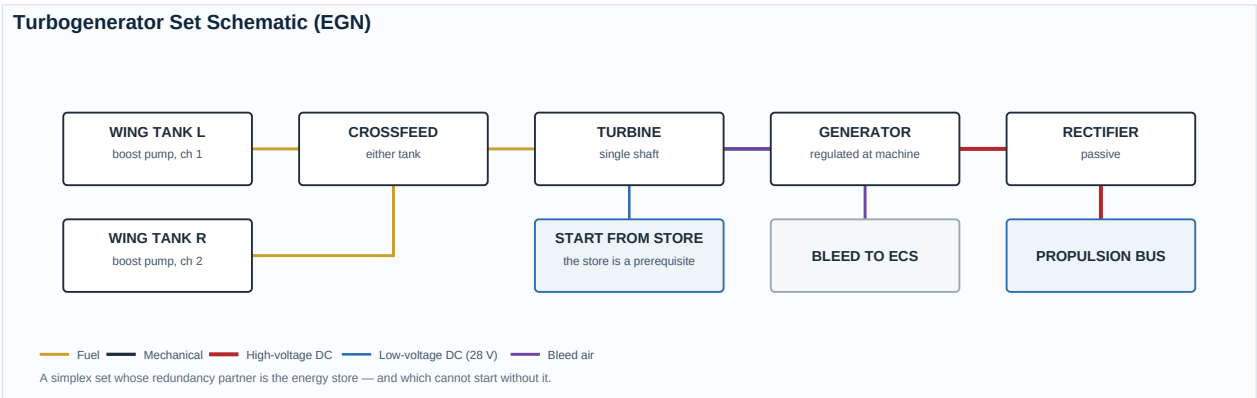


Figure 6-5 Turbogenerator Set (EGN). The turbine, gearbox, generator and rectifier in the aft fuselage, with the fuel gallery arriving from the wing tanks on route R-3 and the firewall shutoff valve marked outside the fire zone. The start path from the energy store is drawn as a separate line. Read off it: the generator cannot start without the store.

6.5 Thermal Management (TMS)

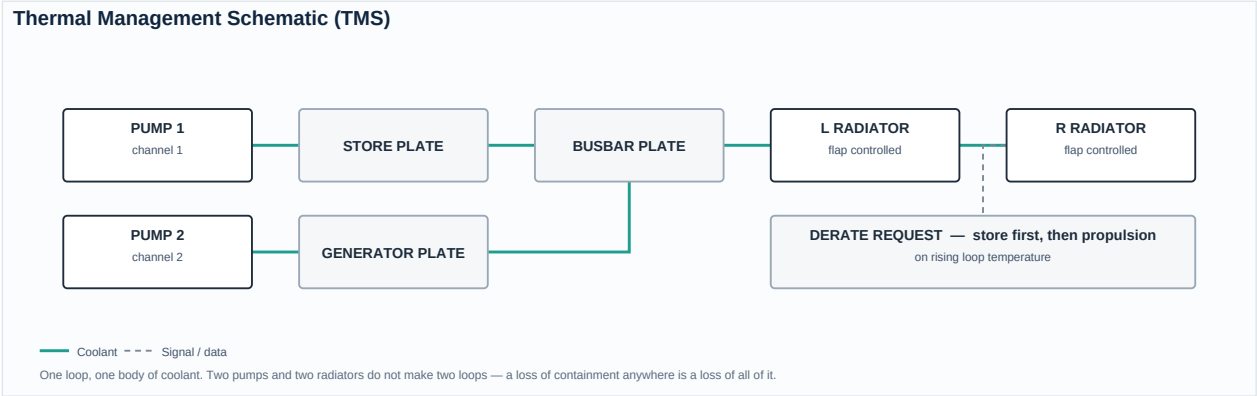


Figure 6-6 Thermal Management (TMS). The coolant ring as a closed loop: two pumps in the aft fuselage, the store and busbar cold plates, both nacelle radiators with their flaps, and the generator plate — every heat source on one circuit. The single bulkhead penetration at frame 14 is marked. Read off it: there is one body of coolant, and the two pumps do not make two loops.

6.6 Flight Control Channel A (FCA)

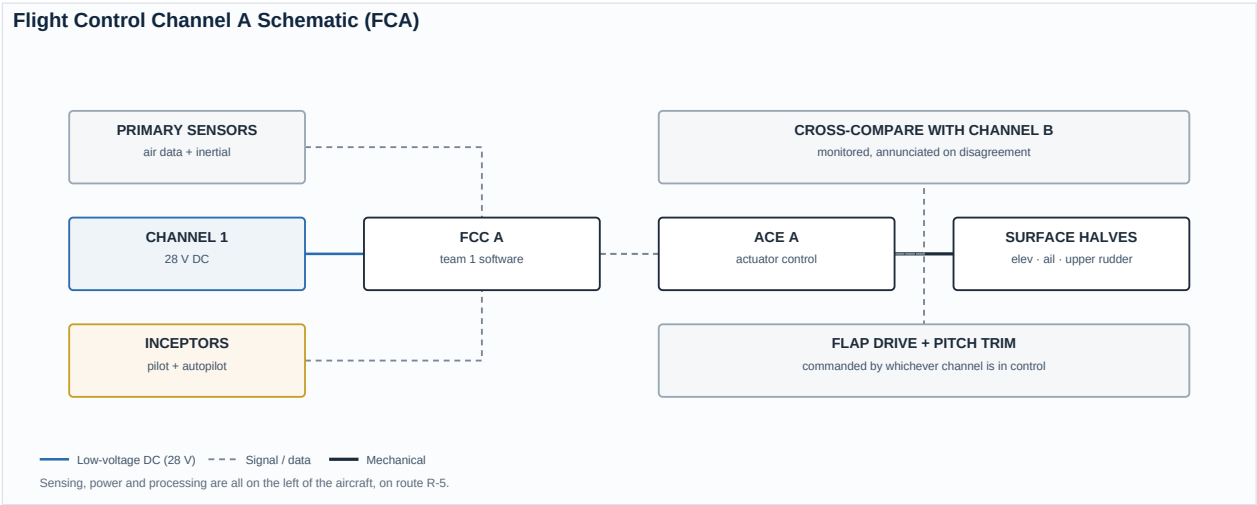


Figure 6-7 Flight Control Channel A (FCA). The channel A computer in the forward rack, its actuator control electronics, and the actuators it drives — one half of each surface pair — with its air-data source shown coming from the primary sensor set. The channel-1 electrical feed follows route R-5 throughout. Read off it: the channel's sensing, power and processing are all on the left of the aircraft.

6.7 Flight Control Channel B (FCB)

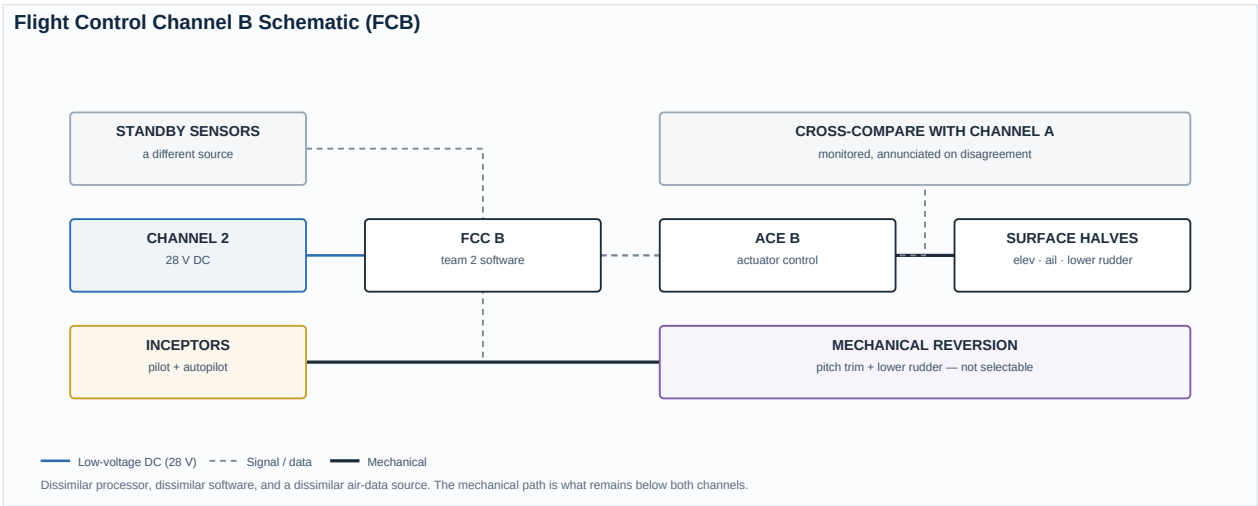


Figure 6-8 Flight Control Channel B (FCB). The channel B computer in the aft cabin bay, drawn to the same key, on route R-6 and channel 2, with its air-data source coming from the standby display. The mechanical reversion path to pitch trim and the lower rudder segment is overlaid as a dashed line. Read off it: the two channels share the airframe and the surfaces, and nothing else.

### 6.8 Electrical Distribution Channel 1 (EL1)

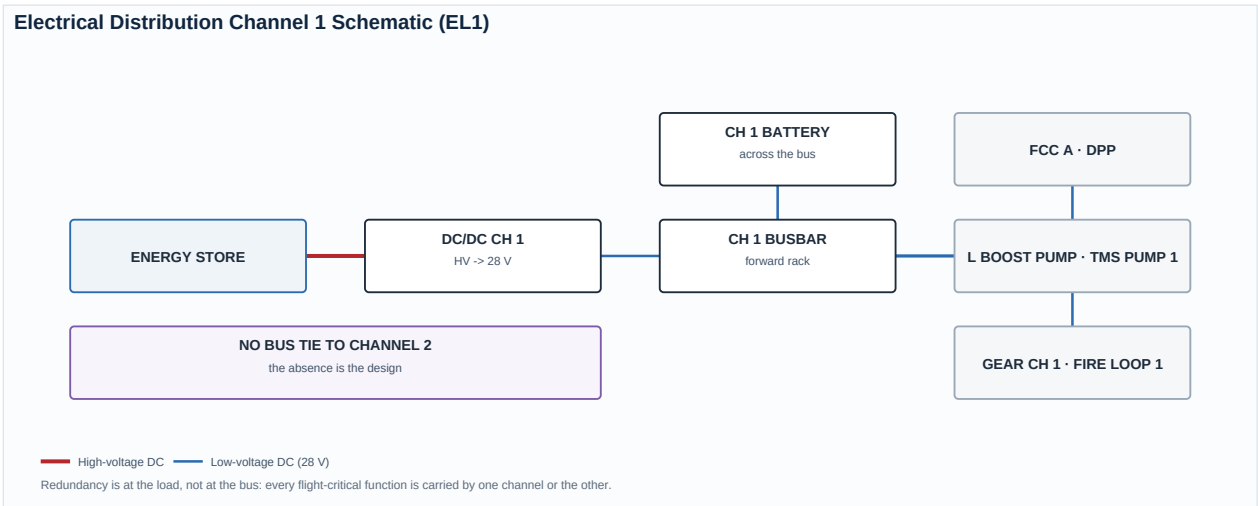


Figure 6-9 Electrical Distribution Channel 1 (EL1). The channel-1 DC/DC converter at the store, the forward busbar and battery, and the solid-state power controllers with their loads named. Read off it: there is no bus tie to channel 2 — the absence is drawn as a gap, not left to inference.

### 6.9 Electrical Distribution Channel 2 (EL2)

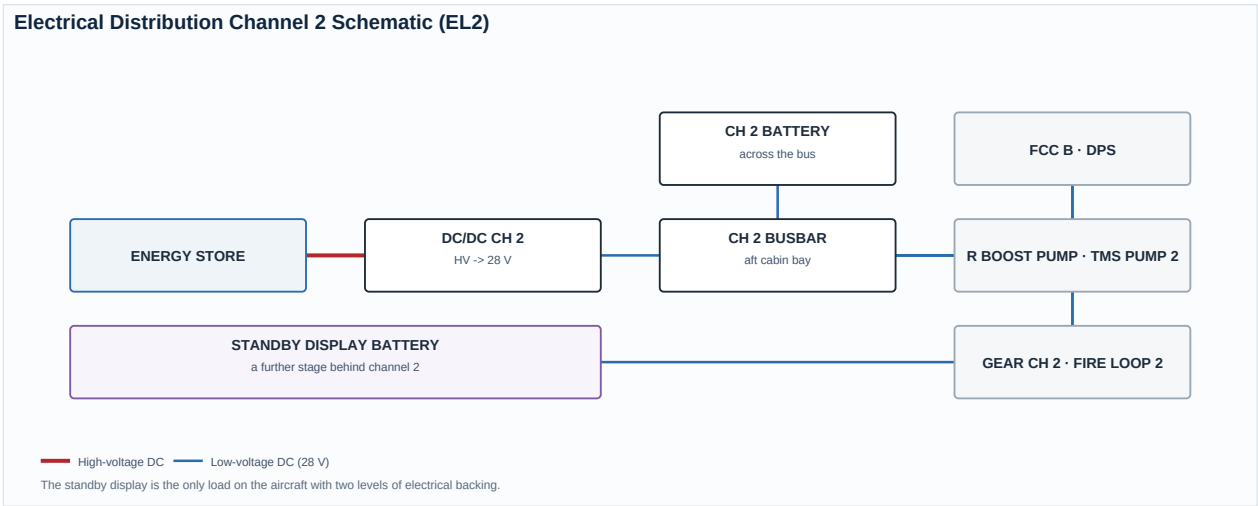


Figure 6-10 Electrical Distribution Channel 2 (EL2). The channel-2 equivalent, with its busbar and battery in the aft cabin bay and the standby display's dedicated battery shown as a further stage behind it. Read off it: the standby display is the only load with two levels of electrical backing.

### 6.10 Primary Flight Display (DPP)

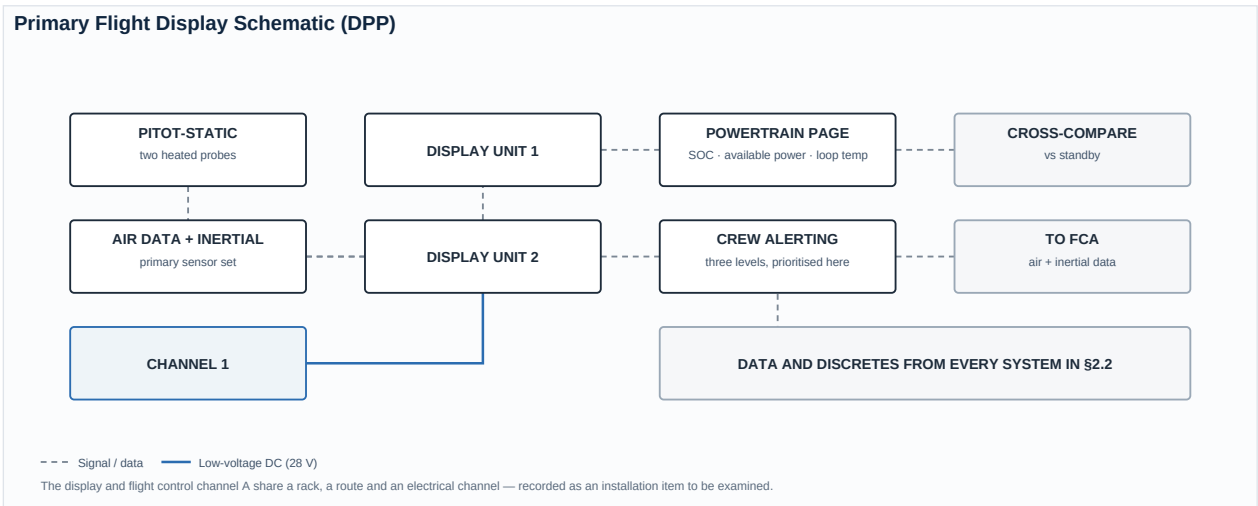


Figure 6-11 Primary Flight Display (DPP). The two display units, the primary sensor set, the two heated probes, and the data paths out to flight control channel A. The crew alerting function is drawn as a block inside the display units with its inputs listed. Read off it: the primary display and channel A share a rack, a route and an electrical channel.

### 6.11 Standby Display (DPS)

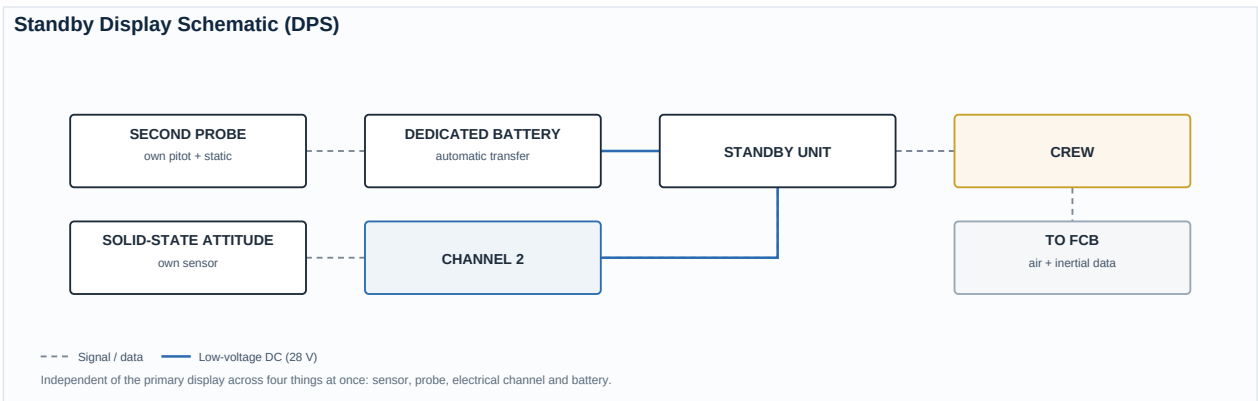


Figure 6-12 Standby Display (DPS). The standby unit with its own sensor, its own probe, its channel-2 feed and its dedicated battery, and its data path out to flight control channel B. Read off it: the independence from the primary display is

across four things at once — sensor, probe, channel and battery.

6.12 Forward Hull Compartment (HUF)

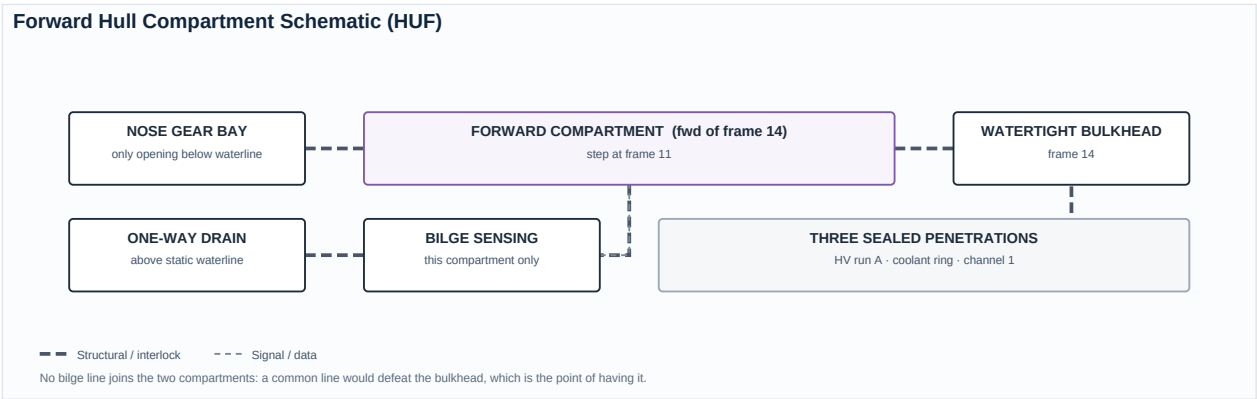


Figure 6-13 Forward Hull Compartment (HUF). Profile and plan of the hull forward of frame 14, showing the step at frame 11, the nose gear bay and its drain, the bilge sensing, and the three sealed penetrations of the bulkhead. Read off it: the bay doors are the only opening below the waterline.

6.13 Aft Hull Compartment & Water Systems (HUA)

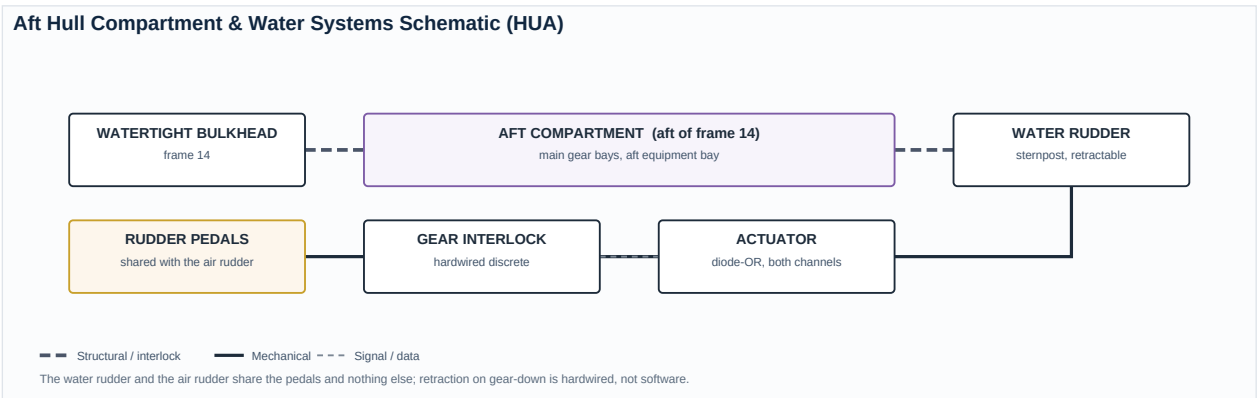


Figure 6-14 Aft Hull Compartment & Water Systems (HUA). Profile and plan aft of frame 14, showing the main gear bays, the water rudder at the sternpost with its actuator above the static waterline, and the mechanical run to the pedals on the opposite side of the hull to the air rudder run. The gear-selection interlock is drawn as a hardwired discrete. Read off it: the water rudder and the air rudder share the pedals and nothing else.

6.14 Amphibious Landing Gear (LDG)

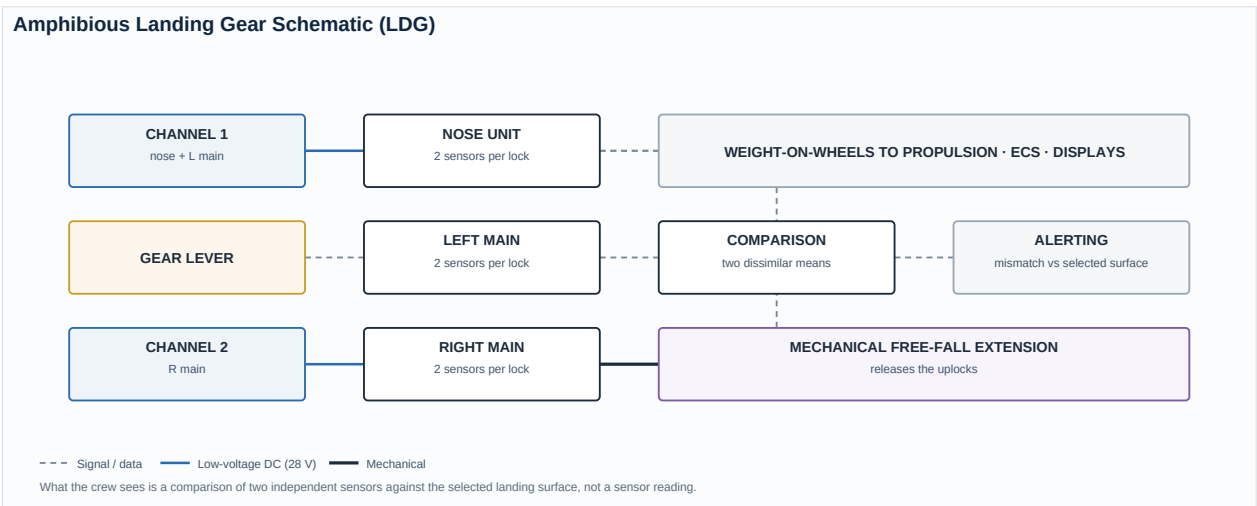


Figure 6-15 Amphibious Landing Gear (LDG). The three units with their actuators, uplocks and downlocks, and the two independent position sensors per lock drawn as separate lines to the alerting function. The channel split — nose and left main on channel 1, right main on channel 2 — is shown at the power inputs. Read off it: the indication the crew sees is a comparison, not a sensor.

6.15 Fire & Thermal Protection (FPR)

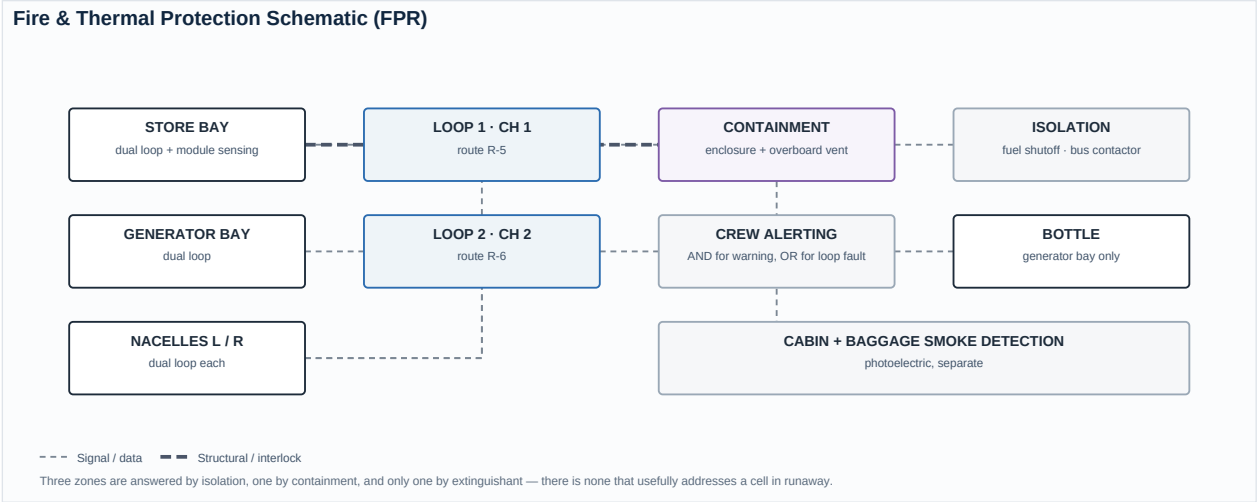


Figure 6-16 Fire & Thermal Protection (FPR). The four detection zones with their dual loops on separate channels, the store bay's additional module-level sensing, the single extinguishant bottle in the generator compartment, and the isolation commands out to the fuel shutoff and the bus contactors. Read off it: three zones are answered by isolation, one by containment, and only one by extinguishant.

6.16 Cabin Environment (ECS)

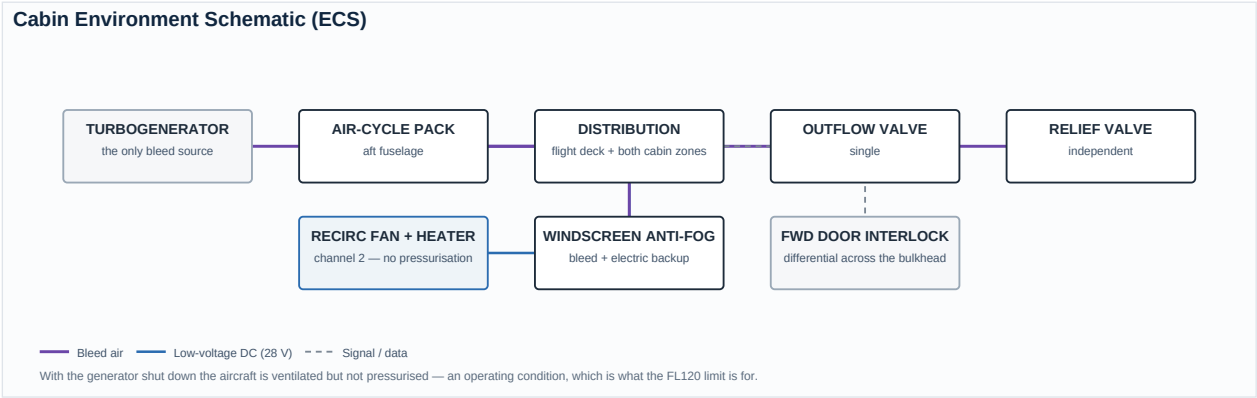


Figure 6-17 Cabin Environment (ECS). The bleed offtake from the turbogenerator, the air-cycle pack, the distribution ducting forward, the outflow and relief valves at the aft bulkhead, and the electrical recirculation and heating path shown as the alternative to the pack. The cabin air inlet on the upper fuselage and the store vent on the underside are both marked. Read off it: with the generator shut down the aircraft is ventilated but not pressurised.

## Appendix A — What this document does not do

This document describes the Halcyon HA-10 as designed. It states what the aircraft is made of, how the parts are arranged, what each part needs from the others, where each part is installed, and how the crew operates it. It does not say what any of that is worth in safety terms.

Specifically, and deliberately, it contains no failure conditions, no severity classifications, no probability objectives, no development assurance levels and no safety requirements. Those are the output of the safety assessment, not an input to it, and a design description that pre-empts them makes the assessment a transcription exercise.

The distinction matters most where it is least visible. Section 5 records, for example, that the two propulsion units share part numbers, that the coolant loop is one loop with two pumps, that the propulsion bus is a single electrical node, that the primary display and flight control channel A share a rack and an electrical channel, and that the cabin air inlet is on the upper fuselage while the energy store vents from the underside. Each of those is a design fact. Whether any of them is a common cause, an independence violation, a single point of failure or a non-issue is a question this document raises and does not answer.

The same applies to the numbers. Where a figure is preliminary it says so. Where a design has a declared limit — a sea state, an altitude, a thermal envelope — the limit is stated as a design characteristic, not as a demonstrated capability, and nothing here should be read as a claim that any objective has been met.

Two further notes for anyone working from this document.

First, the aircraft is fictional. The Halcyon HA-10 is not any company's programme; the configuration is a plausible answer to the stated mission and nothing more, and no number in it is taken from, or intended to describe, any real aircraft.

Second, the description is deliberately uneven in one respect. Some systems are described in more architectural detail than others, because some are more architecturally interesting on this aircraft than others. That unevenness reflects the design, not an omission, and the reader should not infer that a shorter section describes a simpler or a less important system.